How Machine Learning Predicts Employment Trends

A data analysis project using NumPy, Pandas and scikit-learn

Linked to UN Sustainable Development Goal 8: Decent Work and Economic Growth

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Note on data: *the dataset used here is simulated. It is a 16-year, 10-sector panel built to follow widely reported patterns (growth in IT, health and green jobs, decline in agriculture and manufacturing, rising automation, and the 2020 shock). All figures are illustrative and are not official statistics. The method is identical to what would be applied to real labour-force data (for example from a national statistics office, the World Bank or the ILO), and the code can be pointed at such a file.*

1. Problem Understanding

1.1 Problem statement

Governments, universities and employers must decide today how many people to train, where to invest and which jobs will need support, but the labour market changes every year. Traditional forecasting relies on a few economic indicators and straight-line extrapolation. Machine Learning (ML) offers a data-driven alternative: it learns patterns from past employment records, together with signals such as economic growth, automation and skill demand, and uses them to predict where employment is heading.

Question addressed: how does Machine Learning predict employment trends, and how well does it work in practice?

1.2 Objectives

- **Explain** the end-to-end process by which ML turns labour-market data into employment predictions.
- **Build** a working example: clean a sector-level employment dataset and analyse it with NumPy and Pandas.
- **Train and compare** ML models that predict next-year employment growth, testing them honestly on years they have not seen.
- **Forecast** sector employment to 2030 and interpret the results.
- **Link** the findings to SDG 8 and discuss limitations.

1.3 How ML predicts employment trends: the concept

Prediction follows a repeatable pipeline (Figure 1). Each stage has a specific job:

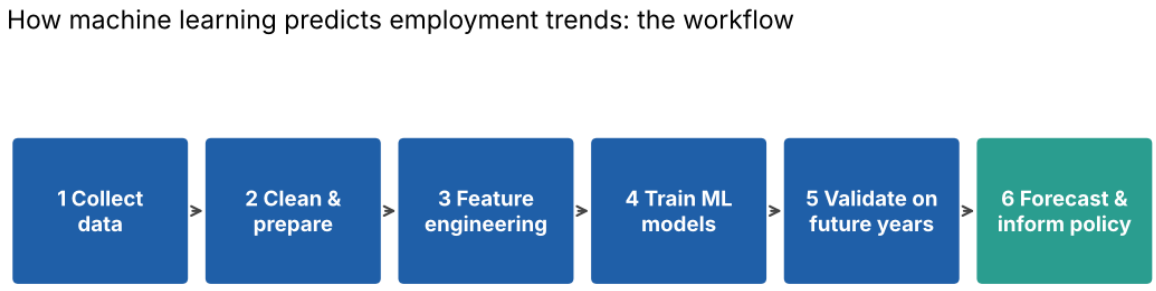


Figure 1: The ML workflow used to predict employment trends

- **Data:** historical employment by sector and year, plus drivers such as GDP growth, job-postings, wages, an automation index and demand for digital skills. Real systems also use online job adverts, surveys and administrative records.
- **Features:** raw data is turned into inputs the model can learn from, for example last year's growth, the change in job postings, or the sector. The key rule is that only information available before the prediction date is used (here, year t-1 predicts year t).
- **Learning:** an algorithm searches for the relationship between the features and the outcome. This is supervised learning, where the outcome (employment growth) is known for past years. Linear regression fits one straight-line relationship; Random Forests and Gradient Boosting combine many decision trees and can capture non-linear effects and interactions, such as automation hurting some sectors much more than others.
- **Validation:** the model is tested on later years it never saw during training. Random shuffling would leak future information, so a time-based split is used.

- **Forecasting:** once validated, the model (or a trend model) projects employment forward. Outputs are used for planning, with the caveat that they are conditional on the past continuing.

2. Data Collection and Preparation

2.1 Dataset

The raw file contains one row per sector per year (2010 to 2025) for ten sectors: IT & Software, Healthcare, Renewable Energy, E-commerce & Logistics, Education, Financial Services, Construction, Manufacturing, Agriculture and Retail. Variables:

Variable	Meaning
employment_m	Employment in the sector, in millions of people
gdp_growth	National GDP growth for the year (%)
automation_index	Index of automation adoption (higher = more automated)
digital_skill_demand	Index of demand for digital skills in job adverts
job_postings_index	Job-postings activity index
wage_growth	Sector wage growth (%)

Real labour data is almost never clean, so quality problems were deliberately included in the raw file to demonstrate the preparation step: missing values, duplicate rows and typing errors.

2.2 Cleaning steps and results

Problem found	Method (Pandas / NumPy)	Result
10 missing values (automation, skills)	groupby(sector).interpolate()	0 missing
3 duplicate rows	drop_duplicates()	3 rows removed
2 impossible values (employment 10x too high, a typing error)	Compare each value to its sector median; flag if more than 5x; replace by interpolation	2 corrected
Sorting and types	sort by sector and year; sector as category	Clean time order
Final shape	163 raw rows to 160 clean rows	160 x 8, no gaps

Interpolating within each sector keeps the time pattern intact, which matters because a global mean would pull a small sector like Renewable Energy towards a large one like Agriculture.

```
df = raw.drop_duplicates().sort_values(["sector","year"])
med = df.groupby("sector")["employment_m"].transform("median")
bad = (df["employment_m"] / med) > 5          # vectorised outlier rule
df.loc[bad, "employment_m"] = np.nan
cols = ["employment_m","automation_index","digital_skill_demand"]
df[cols] = df.groupby("sector")[cols].transform(lambda s:
s.interpolate(limit_direction="both"))
```

3. Analysis using NumPy and Pandas

All calculations use vectorised Pandas and NumPy operations (groupby, pct_change, transform, corr, polyfit) instead of Python loops, which keeps the code short and fast.

3.1 Growth rates by sector

```
df["growth_pct"] = df.groupby("sector")["employment_m"].pct_change() * 100
s = df.groupby("sector").agg(emp_2010=("employment_m","first"),
emp_2025=("employment_m","last"))
s["cagr_pct"] = ((s.emp_2025 / s.emp_2010) ** (1/15) - 1) * 100
```

Sector	2010 (m)	2025 (m)	CAGR %	Net change (m)
Renewable Energy	0.4	1.3	8.3	+0.9
IT & Software	3.0	8.3	7.0	+5.3
E-commerce & Logistics	2.0	4.6	5.7	+2.6
Healthcare	4.5	7.6	3.5	+3.1
Financial Services	3.5	5.2	2.7	+1.7
Construction	12.0	16.5	2.1	+4.5
Education	5.0	6.7	1.9	+1.7
Retail	8.0	9.1	0.8	+1.1
Manufacturing	18.0	16.9	-0.4	-1.1
Agriculture	45.0	34.6	-1.7	-10.4

Total employment across the ten sectors rose from 101.4 million in 2010 to 110.7 million in 2025 (+9.1%). That modest net figure hides a large reshuffle: the four fastest-growing sectors added about 11.9 million jobs while agriculture alone lost 10.4 million.

3.2 Predictive modelling

The target is each sector's employment growth in year t. The inputs are last year's growth, GDP growth, automation index, digital-skill demand, job postings, the change in postings, and the sector. The data was split by time: 100 rows from 2012 to 2021 for training and 40 rows from 2022 to 2025 for testing. Three ML models were compared with a simple baseline that just repeats last year's growth.

```
train, test = d[d.year <= 2021], d[d.year >= 2022]
models = {"Linear Regression": make_pipeline(StandardScaler(), Ridge(1.0)),
          "Random Forest": RandomForestRegressor(300, max_depth=5, random_state=0),
          "Gradient Boosting": GradientBoostingRegressor(200, max_depth=2,
learning_rate=.05)}
for name, m in models.items():
    m.fit(train[X_cols], train["growth_pct"])
    print(name, mean_absolute_error(test.growth_pct, m.predict(test[X_cols])))
```

Model	MAE (percentage points)	R-squared
Naive (last year's growth)	0.54	0.95
Linear Regression	1.81	0.68
Random Forest	0.74	0.91
Gradient Boosting	1.00	0.87

MAE is the average size of the prediction error in percentage points of growth; R-squared is the share of variation explained (1.0 is perfect).

3.3 Forecast to 2030

For the longer-range forecast, a trend model was fitted to each sector's log-employment with NumPy (np.polyfit), which assumes each sector keeps growing at its recent compound rate. Results are shown in Figure 7.

```
coef = np.polyfit(gp.year - 2010, np.log(gp.employment_m), 1)
forecast = np.exp(np.polyval(coef, np.arange(2026, 2031) - 2010))
```

4. Visualization and Interpretation

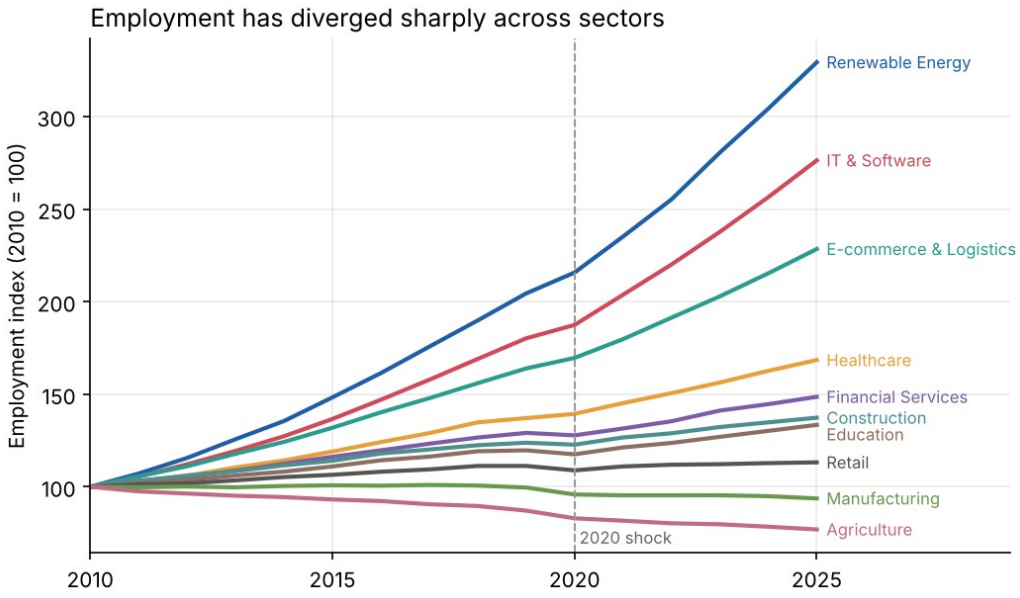


Figure 2: Employment index by sector (2010 = 100)

Interpretation: Renewable Energy more than tripled and IT & Software grew by almost 2.8 times, while Agriculture and Manufacturing ended below their 2010 level. Every sector slows around 2020, then recovers. The labour market is not moving as one block, so a single national forecast would miss the story. Models must be sector-aware.

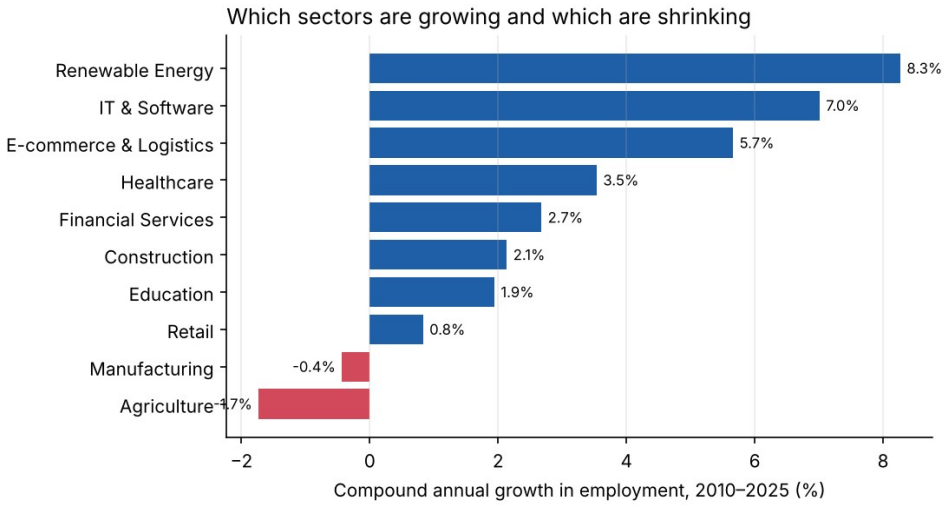


Figure 3: Compound annual employment growth by sector, 2010-2025

Interpretation: Four sectors grow above 3.5% a year, and two shrink. These growth rates are the quantity the ML models try to predict. Note that the shrinking sectors are also the largest employers, so the human impact of decline is bigger than the percentages suggest.



Figure 4: Correlation between employment growth and its possible drivers

Interpretation: Growth is most strongly linked to job postings (0.73) and wage growth (0.93, which moves almost together with employment in this data). Digital-skill demand (0.35) and GDP growth (0.27) show weaker links. Automation shows almost no pooled correlation (0.05) because it rises in every sector and every year, so its effect only shows up when sectors are compared, which is why tree-based models that handle sector differences are useful. Correlation is not causation; these are signals for prediction, not proof of cause.

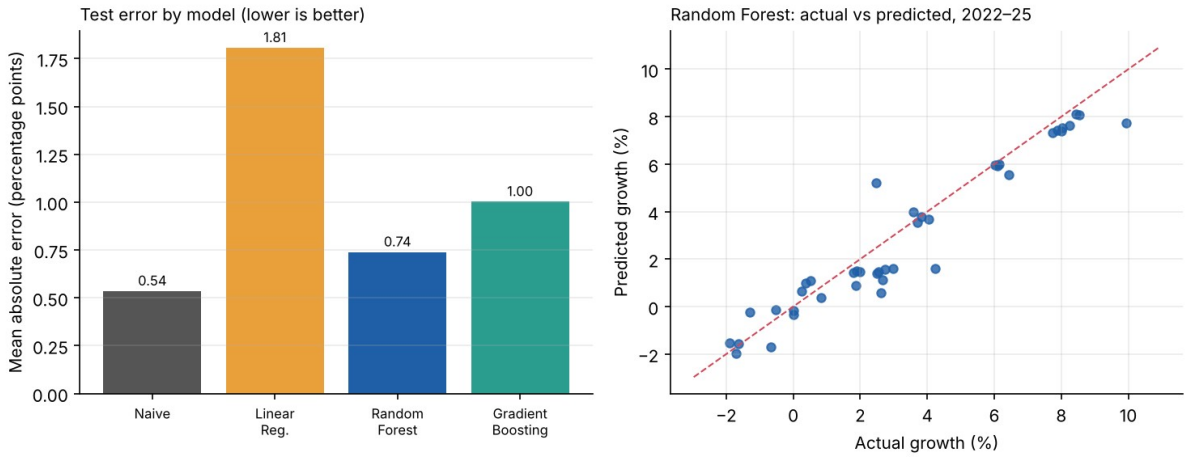


Figure 5: Model comparison and actual versus predicted growth

Interpretation: The Random Forest was the best of the ML models, with predictions clustered near the diagonal line (perfect prediction) and an average miss of 0.74 percentage points. The linear model did worst (1.81), as it cannot capture differences between sectors. One important and honest finding: the naive baseline (repeat last year's growth) scored best (0.54). In this small, smooth dataset growth is highly persistent, so there is little for a complex model to add with only 100 training rows. ML earns its value on larger, noisier data with many more indicators and when conditions change, and a good analyst always checks against a simple baseline as done here.

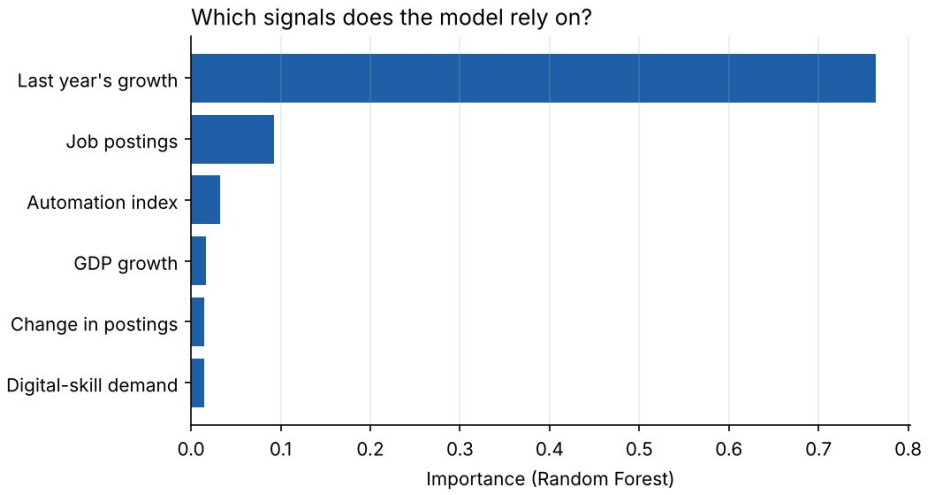


Figure 6: Feature importance in the Random Forest

Interpretation: Last year's growth carries about 76% of the model's importance, followed by job postings (9%) and automation (3%). Momentum dominates, and job postings are the most useful leading signal, which matches how real systems use online vacancy data as an early warning.

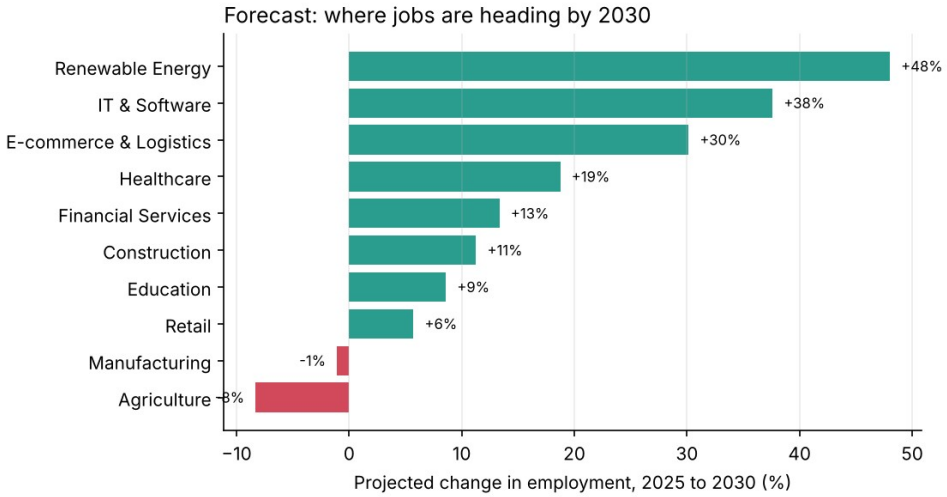


Figure 7: Projected change in employment, 2025 to 2030

Interpretation: If current trends continue, Renewable Energy (+48%), IT & Software (+38%) and E-commerce & Logistics (+30%) lead, while Agriculture (-8%) and Manufacturing (-1%) continue to fall. This points to where reskilling is most urgent: workers leaving agriculture and routine manufacturing need pathways into the growing sectors. Forecasts assume past trends continue, so they are scenarios, not certainties.

5. Innovation and SDG Mapping

5.1 Real-world relevance

Public employment services, education planners and large employers already use ML on job-posting and workforce data to anticipate skill shortages, target training budgets and flag regions at risk of job losses. This project reproduces that logic on a small scale and adds two practical ideas: a sector-aware model that separates automation-exposed sectors from growing ones, and a baseline check that stops a complicated model being trusted without evidence.

5.2 Link to the Sustainable Development Goals

SDG	Target	How this work connects
SDG 8: Decent Work and Economic Growth (primary)	Full and productive employment; reduce the share of youth not in employment, education or training	Early prediction of job growth and decline lets policymakers act before unemployment rises, and shows where jobs are being created
SDG 4: Quality Education	Increase the number of people with relevant skills for employment	Forecast of growing sectors and rising digital-skill demand guides which courses to expand
SDG 9: Industry, Innovation and Infrastructure	Upgrade industry; foster innovation	Strong growth in renewables, IT and logistics reflects industrial upgrading
SDG 10: Reduced Inequalities	Inclusive economic participation	Highlights the workers (agriculture, manufacturing) most at risk, so support can be targeted

The strongest connection is to SDG 8. Predicting employment trends is not just a technical exercise: it supports the goal of productive work for all by giving decision-makers advance warning. Responsible use also matters: models should be audited for bias (for example, against women or informal workers who are missing from formal data) and should support, not replace, human judgement.

6. Documentation and Conclusion

6.1 Summary of findings

- **Method:** ML predicts employment trends by learning from historical employment data and its drivers, then testing on unseen years before forecasting.
- **Data:** 163 raw rows were cleaned (missing values, duplicates, errors) into 160 consistent rows.
- **Trends:** renewable energy, IT and e-commerce are expanding quickly, while agriculture and manufacturing decline.
- **Models:** Random Forest was the best ML model (MAE 0.74 points, R-squared 0.91), but a simple persistence baseline (MAE 0.54) was even better on this small dataset.
- **Signals:** past growth and job postings are the most useful predictors.
- **Outlook:** by 2030, the largest gains are projected in renewable energy (+48%) and IT (+38%), and agriculture is projected to fall about 8%.

6.2 Conclusion

Machine Learning is a useful tool for predicting employment trends because it can combine many signals, handle sector differences and be tested objectively on unseen years. This project shows the full process from messy data to a forecast and interpretation. It also shows that ML is not magic: with limited, smooth data a simple baseline can be hard to beat, so results must always be validated and reported honestly. Used carefully, these predictions can support SDG 8 by helping societies prepare workers and institutions for the jobs of the future.

6.3 Limitations and future work

- **Simulated data:** results demonstrate the method, not real-world magnitudes. The next step is to rerun the code on real labour-force data.
- **Small sample:** 160 rows limit what complex models can learn; more years, regions and detailed occupations would help.

- **Trend forecast:** assumes past growth continues and cannot anticipate shocks or policy changes; scenario analysis would be better.
- **Not causal:** models find predictive patterns, not causes.
- **Future work:** add regional data, use text from job adverts, try time-series models such as ARIMA or LSTM, and quantify forecast uncertainty.

6.4 Tools and references

Python 3, NumPy, Pandas, Matplotlib and scikit-learn. General background: scikit-learn user guide (scikit-learn.org); United Nations Sustainable Development Goals (sdgs.un.org); International Labour Organization labour-market statistics (ilostat.ilo.org). The full code is submitted with this report as analysis.py, with the data in employment_raw.csv and employment_clean.csv.